Lu Zhang | VFX721 | Spring 2026 | Houdini Version 21.0.596
Project 1: Procedurally Modeling a Building

The Bowery - User Guide and Breakdown

Renderer: Karma CPU

Resolution: 1920 x 1080

Samples: Primary Samples 9, Min Secondary 1, Max Secondary 9

Average render time: ~5.5 min / frame

Min/Max rays: 1 / 9

Additional settings: Diffuse Limit 1, Reflection Limit 4, Screendoor Samples 4

Number of lights: 2

Geometry complexity: ~47,000 primitives

Brief User's Manual

This tool was designed to procedurally generate an apartment building inspired by The Bowery, located in Savannah, GA. All main controls are located in the apartment geometry node.

Parameter Controls:



Width — Controls the overall width of the building.

Height — Controls the total height of the building.

Depth — Controls the overall depth of the building.

Floor Height — Controls the height of each individual floor. This affects the spacing and scale of windows and doors on every floor.

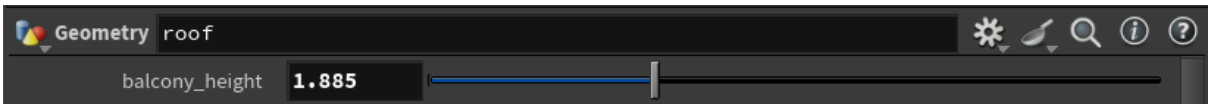
Column Width — Controls the width of each facade column module. Adjusting this will affect how wide each window bay appears.

Column Gap — Controls the spacing between each column module along the facade.

Column Side Door Width — Controls the width of the side door element within each column.

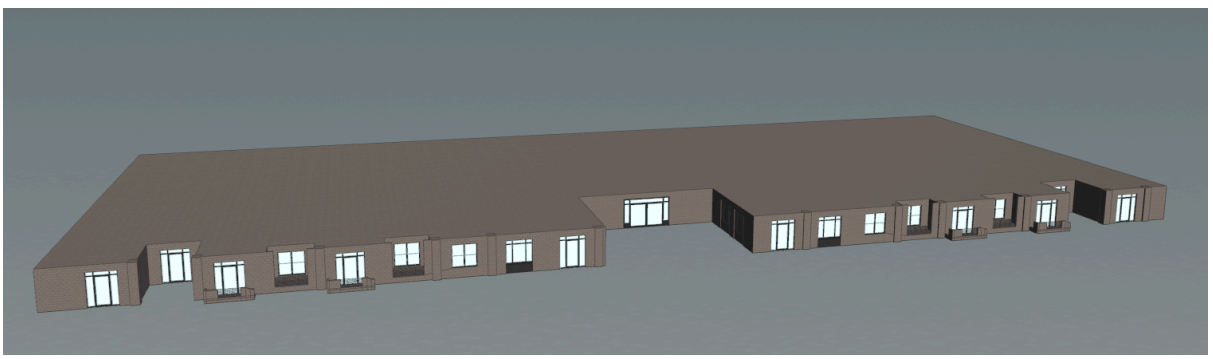
Column Window Width — Controls the width of the windows within each column module.

The roof geometry node contains one additional parameter:



Balcony Height — Controls the height of the balcony parapet on the roof level.

First floor:



Middle floor:



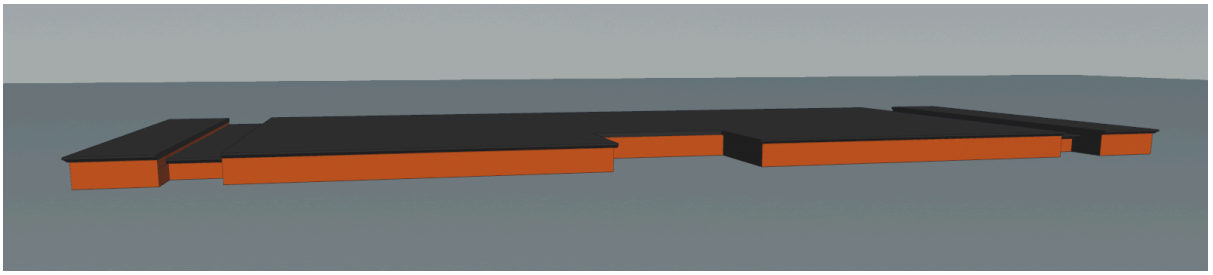
Last floor:



Combined body part:



Roof:



Combined:



Technical guide

All geometry is contained within a single apartment geometry node, organized into separate network boxes for the building body and roof to keep the network clean and accessible.

Procedural Element Placement:

The placement of facade elements such as windows, doors, and columns was managed through a Line → Delete → Copy to Points workflow. A line is first generated based on the building's width or depth parameter, then points are deleted or kept using filter expressions to define placement positions. Geometry is then copied onto these points. This approach allows the number of elements to automatically respond to changes in width and depth without manual intervention.

Floor Generation with Foreach Loop:

A foreach loop was used to procedurally stack floors along the Y axis. Rather than duplicating node networks for each floor type, if statements written directly into parameter expressions are used to detect the current iteration number and control the geometry generated at each step. This allows for meaningful differences between the ground floor, standard middle floors, and the top floor — such as variations in window placement, door elements, and balcony details — all handled within a single compact network.

Material Assignment:

Materials were assigned using a Name attribute workflow. Each component was given a unique name attribute at the time of creation using a Name SOP. After all geometry was merged together, a Groups from Name SOP was used to automatically convert the name attributes into primitive groups. Material SOPs were then applied to each group to assign the correct material. This approach is efficient because material assignments are embedded directly into the geometry as it is built, requiring no manual selection or re-grouping after the final merge.

Description of What Has Been Done Beyond the Requirements

Extra Parameters：

Beyond the required width, depth, and height, additional parameters were added to give the user more precise control over the building's appearance. These include floor height, column width, column gap, column side door width, and column window

width — all of which affect the facade layout — as well as balcony height for the roof level.

Texturing:

A combination of downloaded textures and hand-painted textures were used for the building's facade materials to closely match the reference building, The Bowery in Savannah, GA. Multiple material assignments were made across different parts of the building including walls, windows, door frames, and balconies.

BIBLIOGRAPY AND REFERENCES

- www.deborahfowler.com
- https://www.theboweryapartments.com/neighborhood
- https://www.thomasandhutton.com/portfolio_page/the-bowery-savannah-ga/